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A comprehensive survey on contrastive learning

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ABSTRACT

Contrastive Learning is self-supervised representation learning by training a model to differentiate between similar and dissimilar samples. It has been shown to be effective and has gained significant attention in various computer vision and natural language processing tasks. In this paper, we comprehensively and systematically sort out the main ideas, recent developments and application areas of contrastive learning. Specifically, we firstly provide an overview of the research activity of contrastive learning in recent years. Secondly, we describe the basic principles and summarize a universal framework of contrastive learning. Thirdly, we further introduce and discuss the latest advances of each functional component in detail, including data augmentation, positive/negative samples,network structure, and loss function. Finally, we summarize contrastive learning and discuss the challenges, future research trends and development directions in the area of contrastive learning.

1. Introduction

Unsupervised learning [1] refers to the ability of machines to learn effective data representation from unlabeled data. Compared with traditional supervised learning, because unsupervised learning does not require labeled data, it can save a lot of manpower and resources. Contrastive learning is an instance-based [2] unsupervised learning method [3–7]. As one of the most promising techniques in the field, contrastive learning learns common features between similar instances and distinguishes different features among dissimilar instances. In feature space, contrastive learning brings similar instances closer together and pushes dissimilar instances farther apart. This process is driven by the principle that the semantic relationships in the data should be reflected in the learned representations. For a specific instance, contrastive learning does not require knowledge of which class the instance belongs to, it only requires understanding which instances are similar or different.

Instance discrimination, initially proposed in InstDisc [2], is an important proxy task in contrastive learning. Many contrastive learning methods utilize instance discrimination tasks. The core idea of the instance discrimination method is to treat each data sample as an independent instance. During the training process, the goal of the model is to learn a feature mapping function that maps the feature vector of each instance into a unique low-dimensional space, ensuring that different instances can be clearly distinguished in this space. Each image may be significantly different from other images in the same semantic class. While class-supervised learning maintains intra-class similarities, if a model can learn to distinguish individual instances, without any notion of semantic categories, it may result in a representation that captures the apparent similarity between instances [8]. This method not only enhances the discriminative power of the model but also contributes to the robustness of learned features across various tasks.

Significant progress has been made in contrastive learning pretraining methods based on instance discrimination proxy tasks, and unsupervised pretraining efforts have often matched and sometimes even exceeded supervised methods in terms of fine-tuning downstream tasks. In 2021, Zhao et al. [9] explored what makes pre-training based on instance discrimination tasks beneficial for transfer learning, and what knowledge is learned and transferred from these models. The experimental results showed that pre-trained networks on instance discrimination tasks effectively transfer low-level and mid-level representations, but not high-level representations. This indicates that while such networks capture essential features for image recognition, they may lack the abstract semantic understanding typically provided by high-level representations. Meanwhile, the contrastive models learned by the instance discrimination proxy task encapsulate substantial information necessary to reconstruct pixels from output features. This finding means that while the network learns to distinguish between different instances, it also implicitly learns an overall encoding of the entire image.

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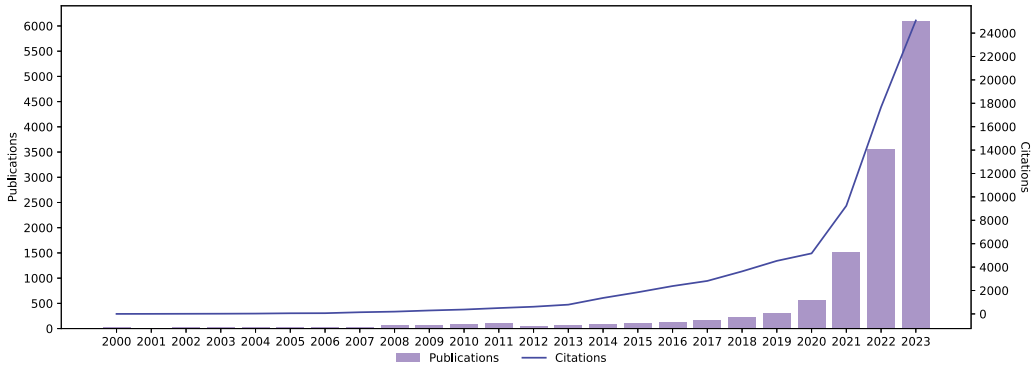

Fig. 1. The histogram of the number of published literature and the number of citations from 2000 to 2023.

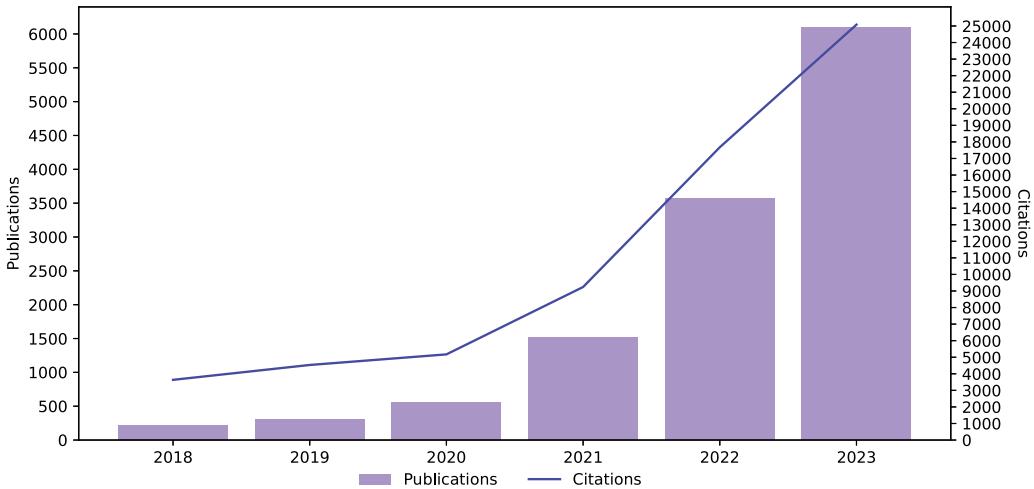

Fig. 2. Histogram of the number of published literature and the number of citations in 2018–2023.

However, handling positive and negative samples in contrastive learning is still a significant challenge, and much work [10–14] has been dedicated to this area (see Section 3.2 for details).

According to the statistics of Web of Science [15], from 2000 to 2023 (statistics until December 31, 2023), the trend in the number of publications on contrastive learning is very pronounced. Using 2018 as the dividing line, the number of publications and citations before 2018 were small. Since 2018, the number of publications and citations in the field of contrastive learning has increased sharply, as shown in Fig. 1. This surge indicates the growing recognition of contrastive learning's importance and its superior performance in various applications. From 2018 to 2023 (statistics until December 31, 2023), the number of publications in the field of contrastive learning reached 12,242, the number of citation articles reached 60,143, and the number of citation articles excluding self-citations remained substantial. The frequency of citations reached 82,186 times, and the average number of citations per article reached about 11, highlighting the field's significant impact and widespread influence.

Fig. 2 shows the trend in the form of a histogram. Fig. 3 illustrates the statistical information about the top four fields with the largest number of published literature on contrastive learning, namely Computer Science, Communication, Engineering, and Mathematics, which account for 84%, 39%, 27%, and 23%, respectively. This data reveals that the application field of contrastive learning is relatively wide, with a significant concentration in Computer Science.

In summary, contrastive learning is a method of machine learning, with its main idea to compare different instances. The purpose of this approach is to identify similarities and differences between instances and use these differences to classify instances. Most contrastive learning methods fall under self-supervised learning, with only a small proportion being supervised learning, such as [16]. Therefore, unless otherwise specified, in this paper, contrastive learning denotes unsupervised contrastive learning. The applications of contrastive learning are diverse and span across multiple domains, such as image recognition [17], image generation [18], adversarial samples [19], video processing [20–26], graph processing [27–32], speech recognition [33], natural language processing [34–38] and recommendation systems, etc. In image recognition, contrastive learning is applicable to image classification and fine-grained classification [39,40], object detection [41], and other tasks. In speech recognition, contrastive learning is useful in speech signal processing and speech recognition. In natural language processing, contrastive learning is utilized in tasks such as text classification and machine translation. In recommender systems, contrastive learning is employed for user behavior analysis and similarity measurement. In conclusion, contrastive learning plays a crucial role in enhancing performance across various fields.

This paper is organized as follows: Section 2 introduces the fundamental principles of contrastive learning, providing a comprehensive overview of the theoretical underpinnings that drive this approach. Section 3 delves into the intricate details of each functional component, including the strategies for data augmentation, the selection and utilization of positive and negative samples, the design and architecture of the neural network, and the formulation of the loss function. Section 4 discusses the various challenges faced in the implementation and optimization of contrastive learning, as well as its diverse applications across different domains. Finally, Section 5 summarizes the key findings and contributions of this work, offering concluding remarks and potential directions for future research.

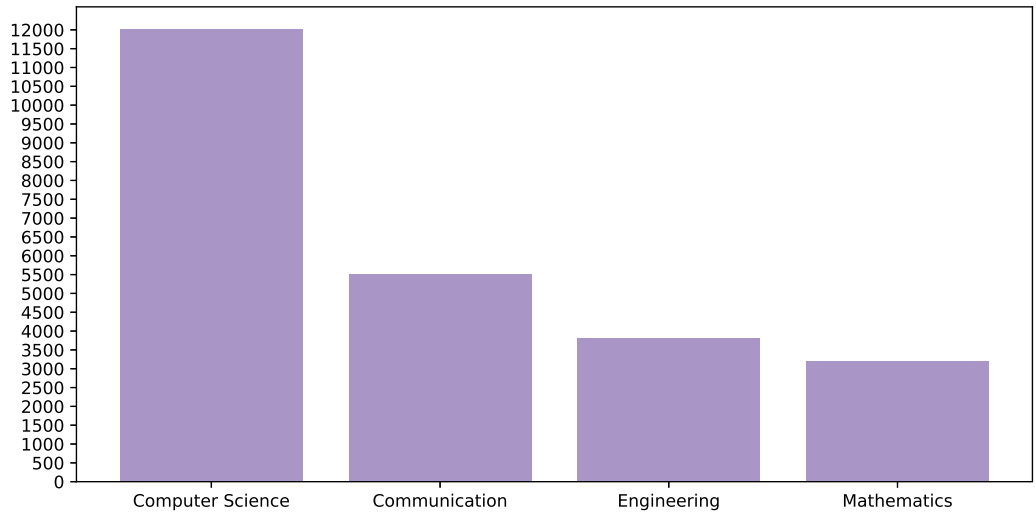

Fig. 3. Top four fields with the largest number of published literature.

2. Principles of contrastive learning

When a child is allowed to observe cats and dogs, even if the child does not know the specific types of cats and dogs, he can learn to distinguish their appearance characteristics, such as ears, noses and tails, thereby finding that cats and dogs are different. Similarly, contrastive learning is like teaching a machine to distinguish objects. It also provides multiple sets of examples to enable the model to learn the similarities and dissimilarities between instances and extract useful features from them to distinguish different instances.

2.1. Basic pipeline

The basic principle of contrastive learning is to learn a model by comparing different examples, where examples can be images, texts, sounds, etc. The goal of contrastive learning is to enable models to distinguish between different instances and to learn to extract useful features from them. In contrastive learning, we usually provide multiple instances, one "similar instance" and multiple "dissimilar instances". Taking images as an example, two random data augmentation strategies are performed on an image to obtain two processed images, one of which is an "anchor point" and the other is a "positive sample", that is, a "similar instance". Since the two images are of the same origin, it is natural to consider the instance as a "similar instance". The other images in the current batch and their images after data augmentation will be regarded as "negative samples", that is, "dissimilar instances". Usually, the more negative samples, the better the effect of the model. When the number of negative samples increases, the model can learn more distinctive features between "dissimilar instances", thereby avoiding excessive attention to the small differences between similar instances and avoiding model over-fitting. This can improve the generalization ability and robustness of the model so that the model can perform better on new instances.

After selecting suitable instances, the learning model will map these instances into the feature space through the encoder and the projection head. The encoder converts the input data into a high-dimensional vector representation and can be trained end-to-end using the backpropagation algorithm. For a pair of similar instances, their features should be similar, thus, the encoder needs to map them into a similar feature space. On the contrary, for dissimilar instances, their features should be dissimilar, thus, the encoder needs to map them to a dissimilar feature space. The projection head maps the high-dimensional vector output by the encoder to a low-dimensional space and optimizes the distance of the embedding vector in the feature space. This process is usually implemented using a fully connected layer. Moreover, a contrastive loss is computed on these feature vectors, and it plays a crucial role in contrastive learning, which is the key to guiding the model to learn the difference between similar instances and the similarity between dissimilar instances. Specifically, the contrastive loss can help the model learn the similarity between anchors and positive samples, and the difference between anchors and negative samples. The overall process is shown in Formula (1).

$$Z = G(F(aug(X)) \tag{1}$$

where X and Z are the input image and the final feature representation, respectively. $aug(\cdot)$ is the random data augmentation method. $F(\cdot)$ and $G(\cdot)$ are the encoder and the projection head, respectively.

2.2. Universal framework

The main framework of contrastive learning is shown in Fig. 4, and can be summarized into three parts, namely, the data processing part, the feature extraction part, and the contrastive loss calculation part. Firstly, the data is processed, which performs data augmentation on the image to form positive and negative samples. Then the image is input into the encoder and the projection head for feature extraction and mapping to obtain the feature representation. Finally, the loss function of the feature representation is calculated to optimize the model.

We have classified some classical contrastive learning methods according to their encoders, whether the network structure is symmetric or not, whether they use negative samples or not, and the loss function, as shown in Table 1.

3. Functional components

3.1. Data augmentation

3.1.1. Conventional data augmentation methods

Conventional data augmentation methods [53–55] include techniques such as cropping, color modification, rotation [56], flipping, noise addition, and blurring. Data augmentation is the primary approach for generating positive and negative samples in contrastive learning. Ge et al. [10] employ texture-based and patch-based data augmentation techniques to generate negative samples. These augmentation strategies are crucial not only for improving model performance but also for significantly influencing downstream tasks [57]. Table 2 presents a comparative analysis of various contrastive learning methods, highlighting the specific data augmentation techniques employed by each method and their respective Top-1 accuracy.

Table 1
Contrastive learning methods statistics.

Method	Encoder	Frame	Negative samples	Loss function
InstDisc [2]	ResNet	Non-symmetrical	✓	NCE Loss
InvaSpread [42]	ResNet	Symmetrical	✓	NCE Loss
CPC [43]	ResNet	Non-symmetrical	✓	InfoNCE
SimCLR [44]	ResNet	Symmetrical	✓	NT-Xent
MoCo [45]	ResNet	Symmetrical	✓	InfoNCE
PCL [46]	ResNet	Non-symmetrical	✓	ProtoNCE
SwAV [47]	ResNet	Symmetrical	✓	CrossEntropy
InterCLR [48]	ResNet	Symmetrical	✓	InfoNCE
BYOL [49]	ResNet	Non-symmetrical	✗	MSE Loss
SimSiam [50]	ResNet	Non-symmetrical	✗	Negative cosine similarity
DINO [51]	ViT	Non-symmetrical	✗	CrossEntropy
AD-CLR [52]	GCN	Non-symmetrical	✓	CrossEntropy

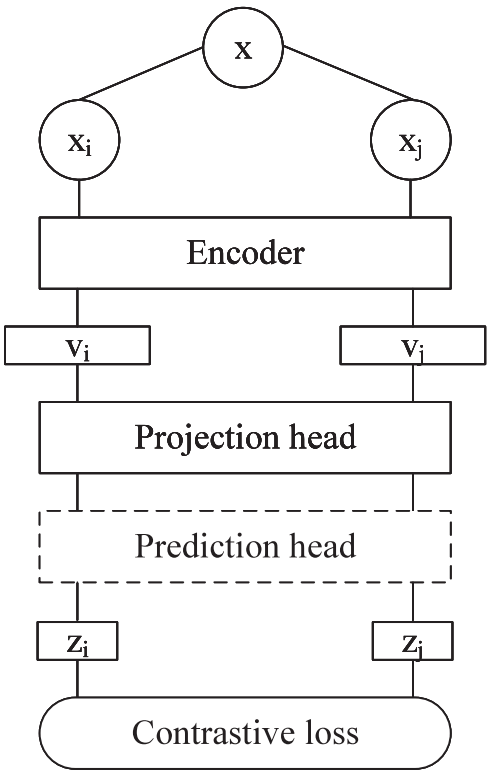

Fig. 4. Summary of Contrastive Learning Frameworks.

InvaSpread [42] can be considered a precursor to SimCLR [44]. Due to its reliance on a limited set of data augmentation techniques, InvaSpread's performance is inferior to that of SimCLR. SimCLR, on the other hand, leverages a diverse array of data augmentation methods, highlighting the importance of augmentation in contrastive learning. It systematically evaluates the impact of various conventional data augmentation techniques to identify those that most significantly enhance model performance, as illustrated in Fig. 5. Where lighter colors of the grid indicate higher accuracy of ImageNet Top-1, and correspondingly, darker colors indicate lower accuracy. Among the methods tested, color modification and cropping emerged as the most effective. The success of SimCLR, achieved through the extensive use of data augmentation, has inspired numerous subsequent studies [58,59] to adopt similar augmentation strategies.

3.1.2. Multi-view data augmentation method

In 2019, Tian et al. [60] proposed a contrastive learning method based on multiview representation learning. The authors argue that an object can be perceived from multiple perspectives in the real world. Each view carries incomplete and potentially noisy information, but the essential features of the object are usually shared between different views. To exploit this, the authors propose a method called Contrastive Multiview Coding (CMC). CMC aims to maximize mutual information [61] between different views corresponding to the same scene, thereby learning a more robust representation of the object. This method also allows for data augmentation by generating diverse views of each training sample, varying in factors such as zoom levels, rotation angles, color, and lighting conditions. The authors demonstrated that as the number of views increases, the performance of the learned feature representation also improves.

In SwAV [47], a new data augmentation method, called the multi-crop strategy, was introduced. This method samples $V+2$ views, consisting of two 224×224 global views and V 96×96 lower-resolution local views. The multi-view approach of multi-crop enables the network to capture a broader range of information from the data, while the inclusion of multiple low-pixel views ensures that the computational load remains manageable.

The advantage of a multi-view approach is its ability to enhance the robustness and generalization capabilities of the model. It allows the model to observe the same sample from different perspectives, thereby learning richer features and information. Thus, the model becomes better equipped to handle challenges posed by deformation, occlusion, and illumination. Additionally, by enhancing data diversity, the multi-view approach can mitigate issues like data bias and overfitting, thereby improving the model's generalization performance.

3.2. Positive and negative samples

3.2.1. Selection of positive and negative samples

In contrastive learning, the selection of positive and negative samples plays a critical role [12,62]. Positive pairs are usually generated using random data augmentations. Because these two images originate from the same source, their features are expected to be close to each other in the feature space. The augmented versions of other images within the same batch are regarded as negative samples [10], and the goal is to maximize the distance between them and the anchor in the feature space.

Traditionally, data augmentation has been the primary method for generating positive and negative samples. However, alternative approaches have also been explored. For example, NNCLR [11] selects the image with the highest similarity to the current image from the support set as a positive sample. T. Park et al. [12] divide images into multiple patches and use different patches as positive and negative samples. SemCL [14] extracts semantic themes from the surrounding context and selects the background portions of the image as negative samples.

Selecting appropriate negative samples proves to be more challenging in contrastive learning than selecting suitable positive samples due to the need to find negatives that are both informative and challenging for the model. For example, Jang et al. [63] classify negative samples into different categories to identify "hard negatives"—those that are close to the anchor in the feature space and semantically similar to the anchor. These hard negatives contain richer information, making them especially valuable for the model to learn from.

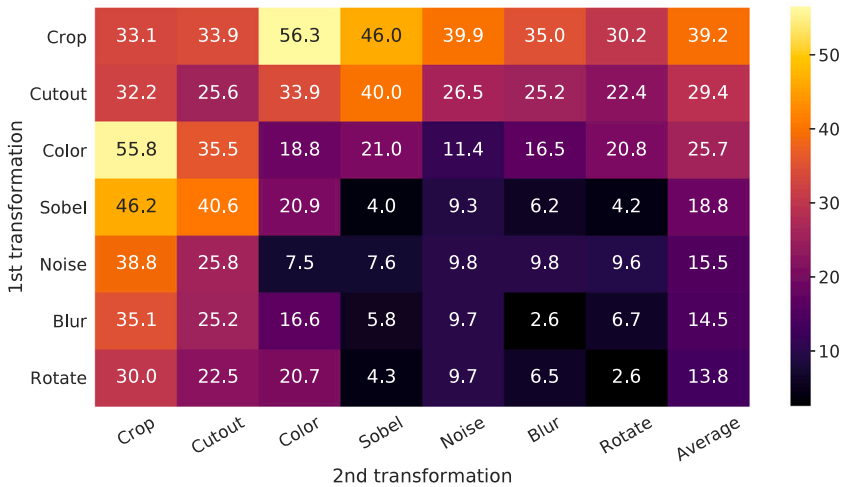

Fig. 5. Experiments on data augmentation in SimCLR [44].

Table 2
Comparison of data augmentation techniques and top-1 accuracy for various contrastive learning methods.

Method	Crop	Cutout	Color	Gray	Flip	Sobel	Noise	Blur	Rotate	Solarization	Multi-view	Top 1
SwAV(NIPS 2020) [47]	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	71.8
MoCo(CVPR 2020) [45]	✓		✓	✓	✓							71.1
SimCLR(PMLR 2020) [44]	✓	✓	✓	✓	✓	✓	✓	✓	✓			69.3
PCL(arXiv 2020) [46]	✓		✓	✓	✓							65.9
SimSiam(CVPR 2021) [50]	✓		✓	✓	✓							71.3
DINO(ICCV 2021) [51]			✓				✓		✓	✓	✓	75.3

3.2.2. *The role of positive and negative samples*

During the training process, the model is tasked not only with bringing positive samples closer to the anchor in feature space but, more crucially, with increasing the distance between the negative samples and the anchor. The role of negative samples is to introduce diversity into the training data, enabling the model to learn both the features of positive samples and the distinguishing characteristics of negative samples, thus enhancing its ability to differentiate between different categories. Negative samples also aid the model in overcoming problems such as data bias and overfitting, thereby improving generalization performance.

In essence, negative samples act as perturbations to the training process, compelling the model to learn more discriminative features and aiding in a deeper understanding of the complexity and diversity inherent in the data. However, in practical applications, the curation of negative samples must consider the specific nature of tasks and datasets to fully leverage their benefits while minimizing unnecessary interference or misleading influences.

3.2.3. *Problems of samples selection*

In contrastive learning, positive samples are typically derived from augmentations of the same source image, while images from different sources are treated as negative samples. However, this approach can lead to certain issues. Junnan Li et al. [46] highlighted that selecting negative samples in this way may inadvertently include images that are semantically similar to the anchor, yet are treated as negatives, thus forcing the model to distance them in the feature space. This misalignment can degrade the model's performance.

To address this issue, methods like PCL [46] and SwAV employ clustering-based techniques [64–69] to first group input data into clusters. These cluster centers are then used as proxies for positive and negative samples, thereby mitigating the problem and enhancing model accuracy.

Compared with traditional contrastive learning methods, SwAV demonstrates greater data efficiency and reduced memory usage by being able to train high-quality visual representations on smaller datasets, requiring only to store a small matrix of visual representations for operation. Additionally, SwAV offers enhanced interpretability by generating cluster assignments that visualize the image representation space, thereby facilitating a better understanding of the visual features learned by the model. This interpretability can provide valuable insights for downstream tasks.

3.2.4. *Solution of samples selection*

In most cases, positive and negative samples are indispensable, with the inclusion of more negative samples generally leading to improved model performance. However, as contrastive learning has evolved, some methods [49,49–51,70,71] have emerged that do not rely on negative samples. These methods achieve feature distinction between instances through techniques such as momentum encoders and gradient stopping. Contrastive learning methods without negative samples, such as BYOL, SimSiam, DINO, and others, tend to require smaller batch sizes. Tian et al. explored why these methods can avoid using negative samples without suffering from feature collapse [72]. Since negative samples are not used, the model's performance becomes less dependent on batch size, with experiments demonstrating that these models are not sensitive to changes in batch size.

3.3. *Network structure*

In contrastive learning, randomly augmented data is typically fed into an encoder to obtain a feature representation. This representation is then passed through a nonlinear layer, which is called the projection head. Finally, the loss function is calculated to optimize the feature space. Earlier contrastive learning methods did not have a relatively fixed network structure but relied on a memory bank to store a large number of negative samples. As contrastive learning has evolved, new structures have been continuously proposed. For example, in MoCo [45], queues and momentum encoders are introduced to generate a large and consistent dictionary that enhances learning efficiency. In SimCLR [44], an MLP nonlinear layer is added after the encoder to form (i.e., the projection head). In BYOL [49], a nonlinear layer

Table 3
Comparison of different encoder components across various contrastive learning methods.

Method	Momentum encoder	Projection head	Prediction head	Top 1
BYOL(NIPS 2020) [49]	✓	✓	✓	74.3
SwAV(NIPS 2020) [47]	✗	✓	✗	71.8
SimCLR(PMLR 2020) [44]	✗	✓	✗	69.3
MoCo(CVPR 2020) [45]	✓	✓	✗	60.6
SimSiam(CVPR 2021) [50]	✗	✓	✓	71.3
DINO(ICCV 2021) [51]	✓	✓	✓	75.3

called the prediction head is added after the projection head to further enhance model performance. With the continuous iteration of network structures, contrastive learning methods have gradually converged on a relatively stable network structure. As shown in Fig. 4, the input data is first augmented and then passed through the encoder, resulting in corresponding feature representations. Finally, the contrastive loss is computed to optimize the feature space for effective learning. Table 3 compares different encoder components used in various contrastive learning methods, focusing on the presence of momentum encoders, projection heads, and prediction heads.

3.3.1. Encoder

In contrastive learning, the encoder is a crucial component of the network structure and plays a central role in processing input data. Traditionally, most contrastive learning methods have employed the classic ResNet-50 as the encoder, leveraging its proven effectiveness. In 2021, the Vision Transformer network [73], based on the transformer architecture [74–77], demonstrated exceptional performance that surpassed ResNet, with the best ViT model achieving 88.55% accuracy on ImageNet1K. As a result, some recent works have adopted ViT as the encoder, such as DINO [51].

3.3.2. Projection head and prediction head

A nonlinear layer, commonly referred to as the projection head, is typically added after the encoder. The concept of the projection head was first introduced in SimCLR and has been shown through extensive experiments to significantly enhance performance. Using a non-linear projection head results in a 3% improvement in accuracy compared to a linear projection and over 10% improvement compared to models without a projection head, highlighting its critical role in the overall model performance. Due to the effectiveness of the MLP layer, the projection head has been adopted by subsequent contrastive learning methods and has become a standard component in contrastive learning frameworks. For example, MoCo v2 [58] incorporates a projection head to enhance its model performance, while a prediction head is added after the projection head in methods like BYOL [49] to achieve even better results, as illustrated in Fig. 6. In the figure, x denotes the input image, and t and t' represent two different data augmentation methods. The augmented images v and v' are processed through the encoders f_θ and f_ξ, followed by the projection heads g_θ and g_ξ to obtain z_θ and z_ξ, respectively. The final z_θ is further processed by the prediction head q_θ to produce the prediction result, with sg indicating the stop gradient operation.

The projection heads discussed above are primarily focused on overall image representations. In contrast, recent research has explored pixel-level contrastive learning [78,79]. For instance, in DenseCL [80], a dense projection head was specifically designed for downstream dense prediction tasks, such as object detection and semantic segmentation. This approach has achieved notable results. In 2022, Wang et al. [81] conducted a comprehensive analysis of the impact of MLP layers on transfer learning performance and concluded that MLP layers play a crucial role in achieving unsupervised learning transfer performance compared to supervised learning.

3.3.3. Momentum encoder

In InstDisc, proposed in 2018, a memory bank is utilized to store a large number of negative samples. Even from a practical standpoint, it is intuitive that incorporating more negative samples generally enhances model performance. The memory bank serves to store these negative samples. However, when dealing with large datasets, using high-dimensional representations for images can result in prohibitive storage costs. Hence, the authors set the dimensionality to 128 to strike a balance between storage efficiency and capacity. This allows the memory bank to accommodate a substantial number of negative samples.

After the encoder processes a batch of images and stores the resulting features in the memory bank, it calculates the loss function and updates the encoder parameters based on this batch of features. Due to parameter updates during the iteration process, there are discrepancies between the features generated from different batches of images. As the number of iterations increases, the discrepancy between features from different batches becomes more pronounced. To address this issue, an effective approach is to allow the encoder to be updated more slowly, thereby ensuring that features remain as consistent as possible across iterations.

In MoCo [45], the concept of a momentum encoder was introduced, which bears certain similarities to knowledge distillation [82–88], as illustrated in Fig. 7. MoCo [45] addresses the storage challenge associated with large batches of negative samples seen in SimCLR [44] by utilizing momentum encoders and queues. The weight function defined in MoCo (see Formula (2)) helps minimize the updates of the encoder parameters [45].

$$\theta_{\mathrm{k}} \leftarrow m\theta_{\mathrm{k}} + (1-m)\theta_{\mathrm{q}} \tag{2}$$

where m is a weight coefficient, typically set to 0.999. In MoCo, the encoder that is updated regularly is denoted as f_q, while the encoder that is updated more slowly is denoted as f_k. θ_q represents the parameters of f_q, and θ_k represents the parameters of f_k. This means that most of the parameters of f_k rely on their previous values, while only a few parameters are updated based on f_q.

3.4. Loss function

A loss function, also referred to as an objective function, is used to measure the discrepancy between the model's predicted results and the actual outcomes during training. In contrastive learning, the loss function guides the feature embeddings such that positive pairs are brought closer together while negative pairs are pushed apart [89,90]. The design of loss functions often depends on the network architecture and the characteristics of the datasets, leading to the proposal of various loss functions tailored to specific scenarios. NCE loss [91] and InfoNCE loss are among the most commonly used loss functions in contrastive learning methods, with their variants, such as symmetric addition and averaging, also being widely adopted. Additionally, other loss functions, including mean squared error (MSE) loss and cross-entropy loss, are sometimes used. Table 4 provides a comparison of various contrastive learning methods, detailing the specific loss functions used and their corresponding Top-1 accuracy. Below, we will provide a detailed introduction to some of the loss functions commonly employed in contrastive learning.

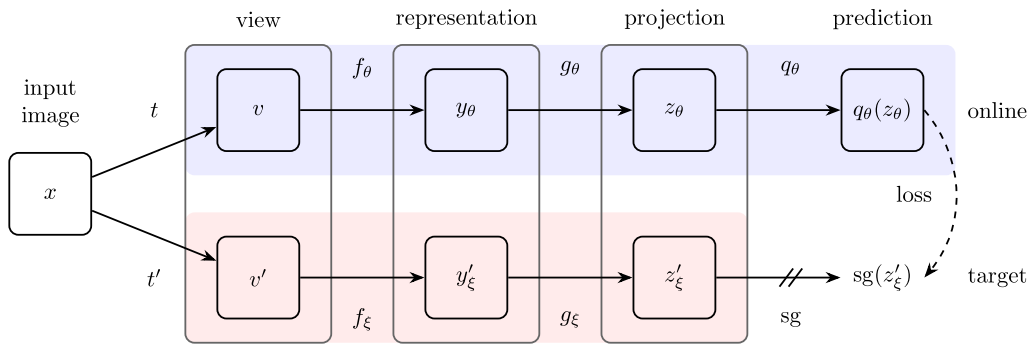

Fig. 6. BYOL network structure diagram [49].

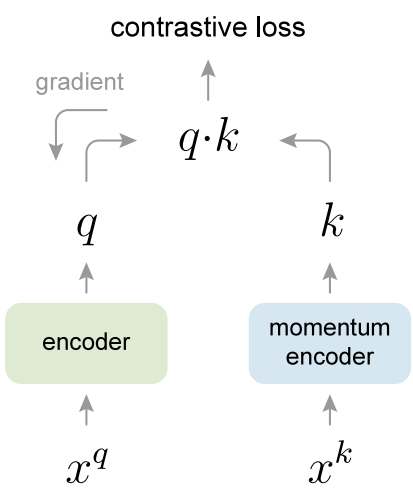

Fig. 7. MoCo network structure diagram [45].

Table 4
Comparison of loss functions and top-1 accuracy for various contrastive learning methods.

Method	Loss function	Top 1
BYOL(NIPS 2020) [49]	MSE Loss	74.3
SwAV(NIPS 2020) [47]	CrossEntropy	71.8
PCL(arXiv 2020) [46]	ProtoNCE	65.9
CPC v2(PMLR 2020)[17]	InfoNCE	63.8
MoCo(CVPR 2020) [45]	InfoNCE	60.6
SimCLR(PMLR 2020) [44]	NT-Xent	69.3
SimSiam(CVPR 2021) [50]	Negative cosine similarity	71.3
DINO(ICCV 2021) [51]	CrossEntropy	75.3

3.4.1. Noise-contrastive estimation(NCE)

NCE loss is very similar to our well-known softmax. The classic softmax calculates the attribution scores for each category of the sample and normalizes them into probabilities. With a large number of categories, the model may become overly conservative in its predictions, assigning relatively uniform probabilities to all categories. In this context, the NCE loss addresses the issue of softmax normalization by transforming the problem into a binary classification task, distinguishing true data from noise samples.

$$\hat{J}_x(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log \frac{p_\theta(y_i|x)}{p_\theta(y_i|x) + kp_n(y|x)} + \frac{k}{m}\sum_{j=1}^{m} \log \frac{p_n(y_j|x)}{p_\theta(y_j|x) + kp_n(y|x)} \tag{3}$$

where k is the number of noise samples. To simplify Formula (3), $n = 1$ and $m = k$, then Formula (3) is rewritten as follows.

$$\hat{J}_x(\theta) = \sum \log \frac{p_\theta(y_i|x)}{p_\theta(y_i|x) + kp_n(y|x)} + \sum_{j=1}^{m} \log \frac{p_n(y_j|x)}{p_\theta(y_j|x) + kp_n(y|x)} \tag{4}$$

In practical applications, the NCE loss is modified and improved into different variants. For example, the loss of variants can perform symmetry processing based on the characteristics of double branches in contrastive learning. Significantly, the symmetric calculation is applied to other loss functions of contrastive learning, such as InfoNCE loss and cross-entropy loss.

3.4.2. Info noise-contrastive estimation(InfoNCE)

The InfoNCE loss function is proposed to further optimize the NCE loss [43], which can maximize the mutual information between positive sample pairs and minimize the mutual information between negative sample pairs. The purpose of contrastive learning is to shorten the distance between similar instances and increase the distance between dissimilar instances. Therefore, the InfoNCE loss is a very commonly used loss function. The InfoNCE loss is illustrated in Formula (5) [45].

$$\mathcal{L}_q = -\log \frac{\exp(q \cdot k + /\tau)}{\sum_{i=0}^{K} \exp(q \cdot k_i/\tau)} \tag{5}$$

where q is the current anchor, $k+$ and k respectively represent positive samples and the number of samples in the current batch, including one positive sample and all other negative samples.

3.4.3. Mean squared error (MSE)

The MSE loss, also called L_2 loss, is the mean value of the distance between instances, as shown in Formula (6). The result of MSE loss is affected not only by positive samples but also by a large number of negative samples, which can evaluate the degree of data change.

$$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - t_i)^2 \tag{6}$$

3.4.4. Cross entropy loss

Cross-entropy loss is mainly used to measure the difference between the predicted distribution of a variable x and the real distribution, and is suitable for classification problems. For a single sample, the cross-entropy loss is defined as follows.

$$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - t_i)^2 \tag{7}$$

Table 5
Top-1 and top-5 accuracies (in %) under linear evaluation on ImageNet.

Method	Architecture	Param(M)	Top 1	Top 5
MoCo [45]	ResNet-50	24	60.6	–
PIRL [8]	ResNet-50	24	63.6	–
CPC v2 [17]	ResNet-50	24	63.8	85.3
PCL [46]	ResNet-50	24	65.9	–
CMC [92]	ResNet-50	47	66.2	87.0
SimCLR [44]	ResNet-50	24	69.3	89.0
MoCo v2 [58]	ResNet-50	28	71.1	–
SimSiam [50]	ResNet-50	24	71.3	–
SimCLR v2 [93]	ResNet-50	24	71.7	–
SwAV [47]	ResNe-50	28	71.8	–
InfoMin Aug. [60]	ResNet-50	24	73.0	91.1
BYOL [49]	ResNet-50	35	74.3	91.6

In contrastive learning, the loss calculation is usually performed in units of a batch of samples, thereby needing some summing and averaging calculations. Thus, Formula (7) is redefined as Formula (8).

$$L = -\frac{1}{batch_size} \sum_{j=1}^{batch_size} \sum_{i=1}^{n} y_i \log \hat{y_{ji}} \tag{8}$$

3.5. Experimental results

Table 5 presents a comprehensive comparison of various contrastive learning methods evaluated under linear evaluation on ImageNet, focusing on their top-1 and top-5 accuracy. It highlights the performance differences between several approaches, all utilizing the ResNet-50 architecture, though with varying numbers of parameters. Among the methods, MoCo achieves the lowest top-1 accuracy at 60.6%, while BYOL attains the highest top-1 accuracy of 74.3%. This indicates the significant advancement in contrastive learning techniques over time. Notably, methods like SimCLR v2 and SwAV also demonstrate strong performance, surpassing the 70% top-1 accuracy threshold, which underscores the effectiveness of more recent innovations in this domain. The top-5 accuracy further corroborates these findings, with methods like InfoMin Aug. and BYOL exceeding 91%. These results emphasize the progress made in self-supervised learning, particularly in the context of contrastive methods, and their growing potential in achieving performance close to supervised learning benchmarks.

4. Discussions

Contrastive learning has emerged as a pivotal paradigm in self-supervised learning, demonstrating its potential to bridge the performance gap with supervised learning approaches. This section delves into the key findings from recent advancements in contrastive learning, highlighting the strengths, limitations, and future directions of this research area.

4.1. Achievements

One of the most significant achievements of contrastive learning is its ability to learn effective representations without requiring labeled data. Methods such as SimCLR, MoCo, and BYOL have shown remarkable improvements in performance on large-scale datasets like ImageNet, often approaching or even surpassing the performance of their supervised counterparts. The introduction of architectural innovations and augmentation strategies, such as multi-crop augmentation in SwAV and momentum contrast in MoCo, have contributed significantly to these advancements. The scalability of contrastive learning methods to large datasets and their robustness to different domains further underscore their utility.

4.2. Challenges

Although contrastive learning has achieved remarkable results in various computer vision and natural language processing tasks, it still faces a series of challenges in its development process.

- For achieving great performance in contrastive learning, a large number of good quality negative samples are needed to train the model. However, how to select high-quality negative samples is a challenge for contrastive learning. If the negative samples are selected as easily distinguishable samples, they do not play a learning role for the model and increase the computational effort for training.
- Contrastive learning faces many challenges on multimodal tasks, which include how to effectively integrate information from different modalities, solve the mismatch problem between modalities, deal with inter-modal heterogeneity, and achieve alignment of cross-modal representations. Meanwhile, the complexity and high dimensionality of multimodal data also increase the difficulty of model training and optimization, and how to effectively utilize these data for model training and inference is also an important challenge.
- In terms of computational resources, contrastive learning requires the solving of complex computational problems, especially on large-scale datasets, and often requires powerful hardware support to meet the high demands of training and inference of contrastive models. These tasks typically require significant computational resources to deal with the complexity and high dimensionality of multimodal data, as well as the large number of parameters and computations required to train models for deep learning, making the acquisition and management of computational resources an important challenge for many researchers.
- In the development of contrastive learning, several methods without negative samples have emerged and they take only positive samples as input. However, this risks the problem of shortcut solutions, which return the same output no matter what inputs are given to the model. Thus it is a challenge to prevent model collapse without negative samples for constraints.
- When contrastive learning is applied to specific domains, integrating and transferring domain knowledge effectively to enhance model generalization and practical application effectiveness becomes a significant challenge. Addressing differences in data features and contextual variations between different domains requires thorough research and effective methodologies to ensure successful applications of contrastive learning in practical scenarios.

4.3. Applications

Contrastive learning is an idea, not a specific model. It performs superiorly on downstream tasks such as image classification, target detection and segmentation. As a result, contrastive learning has an extremely wide range of application areas and uses.

- In the multimodal domain, contrastive learning finds extensive and diverse applications [94–101], spanning various associations such as image-text, audio-text, and image-audio. It is employed not only in tasks like image captioning and speech recognition but also facilitates shared representation learning of multimodal data, fostering cross-modal data mining and the development of multimodal intelligent systems.
- In the medical field, contrastive learning has a wide range of applications [100,102], including medical image analysis, disease diagnosis, drug discovery, and more. Contrastive learning can enhance the accuracy and efficiency of medical image analysis by learning the associations between different structures in medical

images. Additionally, it can learn the associations between disease features and clinical data, assisting doctors in disease diagnosis and prediction.
- In the engineering field, contrastive learning also plays a wide-ranging role [103,104], encompassing various aspects such as engineering image processing, fault diagnosis and prediction, engineering design optimization, and product quality control. By learning the associations between different data, contrastive learning can enhance the accuracy and efficiency of data processing in engineering, providing crucial support and guidance for engineering design, manufacturing, and management.

5. Conclusions

Contrastive learning has gained significant attention due to the ability to avoid the cost of annotating large-scale datasets in various computer vision and natural language processing tasks. In this paper, we systematically review recent advances of contrastive learning from the basic principles, a universal framework to its different functional components including data augmentation, positive/negative samples,network structure, and loss function. Finally, we discuss some solutions to various functional components in contrastive learning.

In the future, the self-supervised contrastive learning will show great potential and a promising application space in the following areas.

- While the current focus is predominantly on images and texts, real-world data often encompasses multiple modalities, such as images, voice, video, and text. Consequently, exploring multi-modal contrastive learning emerges as a promising direction, enabling models to discern correlations among diverse modal data types.
- In terms of computing resources, contrastive learning requires solving complex computational problems, especially on large-scale datasets often requiring powerful hardware. High-performance solutions, such as dedicated devices [105] and strategies such as decoupled hardware [106] designed to accelerate contrastive learning computations, will play a crucial role in overcoming these challenges.
- Beyond its current applications in computer vision and natural language processing, contrastive learning is poised to revolutionize domains such as medical imaging analysis, autonomous systems (e.g., self-driving cars), and personalized recommendation systems. These applications benefit from the ability of contrastive learning to extract rich and discriminative features from complex data sources, thereby improving performance and reliability in critical decision-making tasks.
- As contrastive learning models scale up to handle larger and more diverse datasets, ensuring robust generalization becomes crucial. Future research may focus on developing scalable algorithms that maintain performance across varying data distributions and domains. Techniques like meta-learning and adaptive learning rate scheduling could enhance model adaptability and generalization capabilities.

CRediT authorship contribution statement

Haigen Hu: Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Xiaoyuan Wang:** Writing – original draft, Visualization, Software, Investigation, Data curation, Conceptualization. **Yan Zhang:** Writing – review & editing, Resources, Investigation, Formal analysis, Data curation. **Qi Chen:** Writing – review & editing, Supervision, Software, Resources, Funding acquisition, Conceptualization. **Qiu Guan:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

No data was used for the research described in the article.

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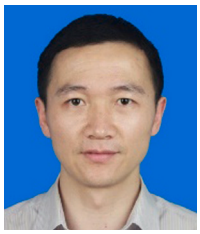

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